

SIH26011 — Quick Technical Reference

3D ULPIN Generation and Vertical Property Mapping

DoLR / Ministry of Rural Development · Software · Smart Automation

Official PS: sih.gov.in/sih2026PS (verified 18 Sep 2026)

Rule: Our 3D ULPIN is a **proposed extension** compatible with existing ULPIN. It is not an official Government format.

WHAT

A working 3D cadastral prototype that:

- takes a **2D land parcel** (and optional LiDAR/DSM + floor plans + utilities),
- builds **legal 3D volumes** for the lot, floors, apartments, common areas, parking, and underground corridors,
- **validates topology** with geometry (not a neural net),
- issues a **proposed 3D ULPIN** nested under the official 14-character surface ULPIN,
- stores LADM-aligned records in PostGIS and shows them on a Cesium globe.

Working name: **ULPIN-3D**. Demo scale: **one Indian urban parcel, one multi-storey building**.

WHY

DoLR: ULPIN is a **14-character** ID from parcel **lat/long vertices**; the ID “would spatially be pointing to the **surface** of the parcel.”

So one official ULPIN cannot uniquely name Flat 501, a basement, an air-rights slab, or a sewer under the same plot. That is the problem SIH26011 asks us to solve.

No public official **3D ULPIN standard** was found (DoLR / DILRMP / NGP 2022). We propose an extension; we do not mint government IDs.

INPUT

SIH portal dataset field is **empty**. We do **not** claim SVAMITVA or state cadastral dumps.

Input	MVP source	Honest note
Cadastral polygon	Authored GeoJSON in EPSG:32643	Format-compatible with GIS; not a live Bhu-Naksha feed
Parent ULPIN	14-char placeholder or ingested string	Labelled official-or-placeholder
Point cloud / DSM	Synthetic LAS or public ALS for metrics	Indian ALS (TALD) only if we actually obtain it
Floor plans	Digitised GeoJSON	Not scanned PDFs in MVP
Utilities	3D line GeoJSON + diameter	Schematic
DEM	Optional; CartoDEM ~30 m	Too coarse for floors — terrain context only
GNSS/CORS	CRS + accuracy metadata	No live CORS client

Missing CRS = **hard fail**. Missing LiDAR or plans = **degraded path**, not a crash.

PIPELINE

Input → CRS/QA → nDSM/footprint → height/floors → extrude solids
→ underground buffers → topology rules → proposed 3D ULPIN
→ PostGIS → API → Cesium

Stage	Tech	Why that tech
I/O, CRS	GDAL, pyproj, PDAL	Industry default; LAS/LAZ + raster
Ground / nDSM	PDAL CSF	No training data
Footprint	Open3D + morph/planarity	Explainable; works on ALS
Floors	Z-histogram / RANSAC, plans win	Airborne LiDAR does not see slabs
Solids	PostGIS SFCGAL CG_Extrude	Real volume + 3D intersection
Topology	SQL + Shapely/GEOS	Deterministic, demo-able errors
IDs	Python issuer	UUID stable; display string hierarchical
API	FastAPI	Same language as the geometry stack
UI	CesiumJS	Globe, underground, WGS84

AI policy: classical geometry is the MVP “automation.” Topology is never ML. Deep models only if time remains, and never as legal truth.

OUTPUT

For each legal space:

- 1. ulpin3d_uuid (immutable)
- 2. Display ID: IN-ULPIN3D/{parent_ulpin}/{CLASS}/{local}/vNN
Example: IN-ULPIN3D/12AB34CD56EF78/UNIT/F05-U501/v01
- 3. 2D footprint + zmin/zmax + 3D prism solid + volume
- 4. geom_origin = SURVEY | PLAN | AI_DERIVED | SYNTHETIC | MANUAL
- 5. confidence 0–1, topology VALID/INVALID, source checksums

Physical building mesh ≠ legal unit. Common corridor is COMMON with undivided share (RERA / apartment acts), not a fake private title.

DEMO (judges, ~4 minutes)

Scene: synthetic **Pune** parcel, real UTM 43N.

- 1. 2D parcel + parent ULPIN
- 2. Load cloud/DSM → extract building → extrude
- 3. Floors → click Flat 501 → cyan volume + 3D ULPIN
- 4. Slice Z → show underground pipe
- 5. Load overlapping fake unit → **INVALID** (red) → fix → **VALID** card

First sentence: *Official ULPIN names the land. We name the volume.*

MVP (must work)

- 1 parcel, 1 building, ≥ 4 storeys, ≥ 2 units/floor, 1 COMMON, 1 parking, 1 utility
- Ingest GeoJSON + LAS/DSM + plans
- Proposed 3D ULPIN under parent ULPIN
- Topology with injected error
- Cesium: isolate floor, underground toggle, provenance
- LADM-aligned schema (not “ISO certified”)

FUTURE (not MVP)

IFC ingest, CityGML export, RandLA-Net, live CORS, city-scale 3D Tiles, real SVAMITVA data, legally binding titles.

RISKS

#	Risk	Mitigation
1	No official Indian LAS	Synthetic georeferenced demo; label sources
2	ALS cannot see floors	Floor plans first; height heuristic flagged
3	ID overclaim	PROPOSED banner; parent ULPIN immutable
4	Dirty polygons break solids	MakeValid + always store 2.5D params
5	DL rabbit hole	Freeze classical pipeline
6	Silent CRS errors	Hard-fail missing EPSG
7	Cesium underground	Translucency; 2D fallback
8	Scope creep	One building
9	Legal overclaim	“Not a title” + geom_origin
10	Unstable IDs	UUID never recycled

JUDGE DEFENSE (short)

Question	Answer
Why not 2D ULPIN?	It points at the surface (DoLR). One XY, many rights.
Why not a building ID?	Building is physical; cadastre is legal space + RRR (ISO 19152).
Official 3D ULPIN?	No. Proposed extension under the 14-char parent.
Common areas?	Undivided share to association (RERA s.17). Class COMMON.
Underground?	Corridor solids + easement-style RRR. Not a new statute.

Question	Answer
Accuracy?	Demo metrics only. Plans ~0.3 m Z. CartoDEM is not floors.
No LiDAR / no plans?	Degraded prisms; confidence drops; no fake VALID.
AI validation?	Topology rules. AI geometry cannot self-certify.
City scale?	Architecture can tile; prototype is one block.
Legal today?	No. Exploration prototype for DoLR.
Standard?	LADM- aligned ; CityGML export later; ULPIN parent per DoLR.

FINAL STACK

Layer	Choice
Language	Python 3.11+
API	FastAPI
DB	PostgreSQL 16 + PostGIS 3.5 + SFCGAL
Cloud	PDAL + Open3D
Vector 2D	Shapely / GEOS / GDAL
Front	React + CesiumJS (+ MapLibre 2D if needed)
IDs	UUID PK + IN-ULPIN3D/... display
Model	LADM-aligned spatial units + RRR
Demo CRS	EPSG:32643 (UTM 43N)

14-day build order: Docker/PostGIS → schema → PDAL nDSM → footprint → floors/plans → extrude → utilities → topology tests → ULPIN API → Cesium happy path → slice/underground/errors → freeze demo.